

Visualizing **Modeled** Networks

a provocation

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Why do we visualize?

- enhanced memory
- simplification
- improved clarity

When do we visualize?

- exploratory
- conceptual
- explanatory

How do we visualize?

- reduce dimensionality
 - ▶ information > ink
 - ▶ signal > noise
- reproducibility
- **retain ethical protections**
- provide interpretive clarity
 - ▶ *theoretical* focus

What do we visualize?

standard approach

- obtain & preprocess
- **visualize** to describe
- analyze
- explain via layering

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proposed approach

- obtain & preprocess
- analyze
- **visualize** to explain

a trivial example

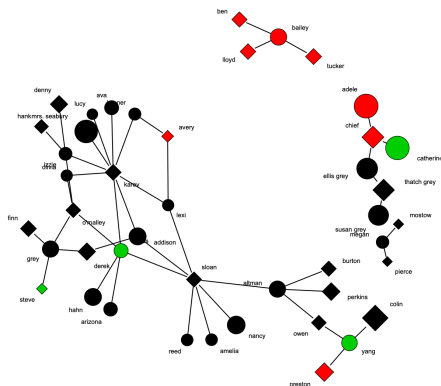


Figure 1: observed data

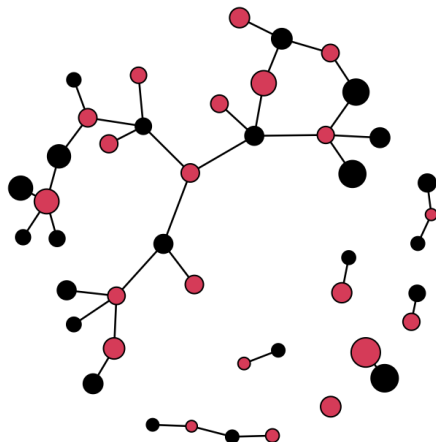


Figure 2: modeled $\sim$ isolates +
gender_diff + age_diff

adapted from [Benjamin Lind](#)

a parallel logic

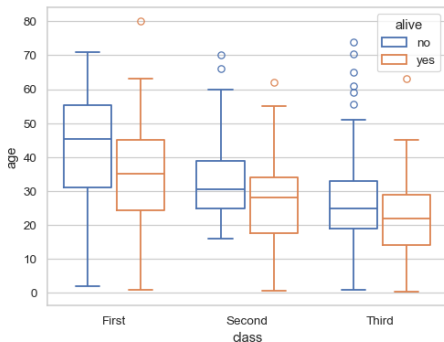


Figure 3: observed data

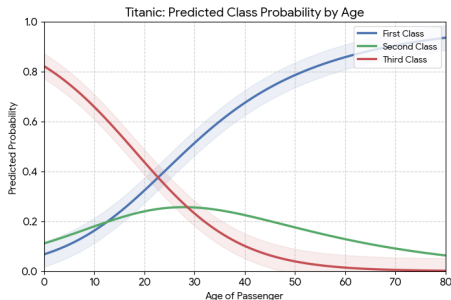


Figure 4: mlogit predicted probabilities

adapted from [Michael Waskom](#)

visualization principles

	observed	modeled
explore/describe	✓	x
simplify	?	✓
reproduce	✓	✓
confidentiality	x	✓
explain/theory	?	✓

the provocation

for some use-cases, presenting **modeled network visualizations** better optimizes our **aims** than observed network data

- focuses on *theoretical* contributions over empirical case idiosyncrasies
- maintains ethical protections
 - ▶ enhanced replicability?
- **NOTE:** this isn't *entirely* new (e.g., block images, motifs)

let's discuss! jimi.adams@sc.edu